

Project: Threat Hunting & Threat Intelligence

by Mallikah Sharp

Introduction

In this project, I will focus on gathering Indicators of Compromise, or IOCs, from open-source threat intelligence platforms. Once I have a set of IOCs like suspicious IP addresses, domain names, or file hashes, I move into the threat hunting phase. For that, I use Azure Microsoft Sentinel and run queries in Kusto Query Language, or KQL. Specifically, I'm querying the *SecurityEvents* table to look for any signs in my environment that match the IOCs I collected. The goal here is to combine real-world threat intelligence with hands-on threat hunting, which gives me practical experience in identifying and responding to potential threats in a security operations context.

Step 1: Gather Threat Intelligence and IOCs

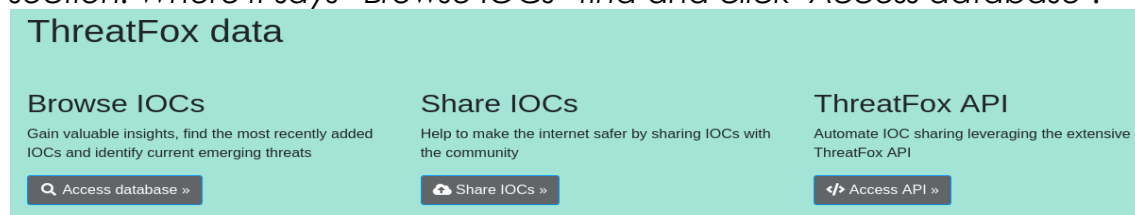
Search for IOCs—such as malicious IP addresses, domains, file hashes, or URLs—from public threat intelligence sources. These indicators can be used to identify potential threats within your environment.

1. Identify Open Threat Intelligence Sources

Use the following websites to gather IOCs:

- **VirusTotal:** A free tool for scanning files, URLs, and IP addresses for malware and other malicious activity.
- **AlienVault OTX:** Open Threat Exchange (OTX) provides crowdsourced threat data.
- **AbuseIPDB:** A database of IP addresses reported for malicious activity.
- **Spamhaus:** Lists of malicious IPs, domains, and email spammers.
- **ThreatFox:** A platform that provides a wide range of IOCs like IPs, domains, and hashes.

***For this project, I will be using the ThreatFox website. Go to <https://threatfox.abuse.ch/> website and scroll down to the ThreatFox data section. Where it says "Browse IOCs" find and click 'Access database'.



The screenshot shows the 'ThreatFox data' section with three main options:

- Browse IOCs**: Gain valuable insights, find the most recently added IOCs and identify current emerging threats. Includes a button 'Access database »'.
- Share IOCs**: Help to make the internet safer by sharing IOCs with the community. Includes a button 'Share IOCs »'.
- ThreatFox API**: Automate IOC sharing leveraging the extensive ThreatFox API. Includes a button 'Access API »'.

***There will be a dashboard with statistics and a list of recently uploaded IOCs.

ThreatFox IOC Database

You are browsing the Indicator Of Compromise (IOC) database of ThreatFox. If you would like to contribute IOCs to the corpuse, you can do so through either the [web form](#) or the [API](#).

 1'417 IOCs shared (past 24 hours)	 Lumma Stealer Most seen malware family (past 24 hours)	 1'409'212 IOCs in corpus
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I will be investigating the Quasar RAT IOC. The Quasar RAT (Remote Access Trojan) is a powerful and open-source malware tool designed to provide unauthorized remote control over Windows systems. Originally developed as a legitimate remote administration tool, it has been widely adopted by cybercriminals and advanced persistent threat (APT) groups due to its extensive capabilities and ease of customization. It was first released in 2014 as xRAT and later renamed to Quasar in 2015, has led to numerous customized variants tailored for malicious purposes. These variants are often used in targeted attacks against government, energy, and education sectors.

2025-05-13 12:01	79.110.49.229:7001		     	 DonPasci
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Quasar RAT offers a comprehensive suite of features that enable attackers to:

- Take screenshots and record webcam activity
- Log keystrokes and capture passwords
- Download, upload, and execute files
- Manage files and processes, including terminating or starting processes
- Open remote desktop sessions and control the mouse and keyboard
- Access system information and edit the Windows registry
- Achieve persistence through scheduled tasks and registry keys

These capabilities make Quasar RAT a potent tool for espionage, data exfiltration, and further malware deployment.

Once I click the link name, I am able to see details about the malware's known aliases, the first and last time it has been seen, the number of IOCs, and a link to malpedia (malware encyclopaedia) where we can look for articles related to this RAT.

Database Entry

Malware:	 Quasar RAT
Malware alias:	CinaRAT, QuasarRAT, Yggdrasil
First seen:	2021-03-12 21:31:00 UTC
Last seen:	2025-05-13 12:01:41 UTC
Number of IOCs:	5'424
Malpedia:	 https://malpedia.caad.fkie.fraunhofer.de/details/win.quasar_rat

2. Collect IOCs

Look for:

- **Malicious IP addresses:** IPs involved in botnets, DDoS attacks, or malicious traffic.
- **Domains:** Known phishing or malware distribution domains.
- **File hashes (MD5, SHA256):** Hashes of malicious files detected by the threat intelligence sources.
- **URLs:** URLs associated with phishing sites or malicious content.

Scroll down the page to see all the IOCs uploaded on the ThreatFox website. It consists of suspicious IP addresses, domains, and File hashes. Choose 5 IOCs to check for them in Microsoft Sentinel logs.

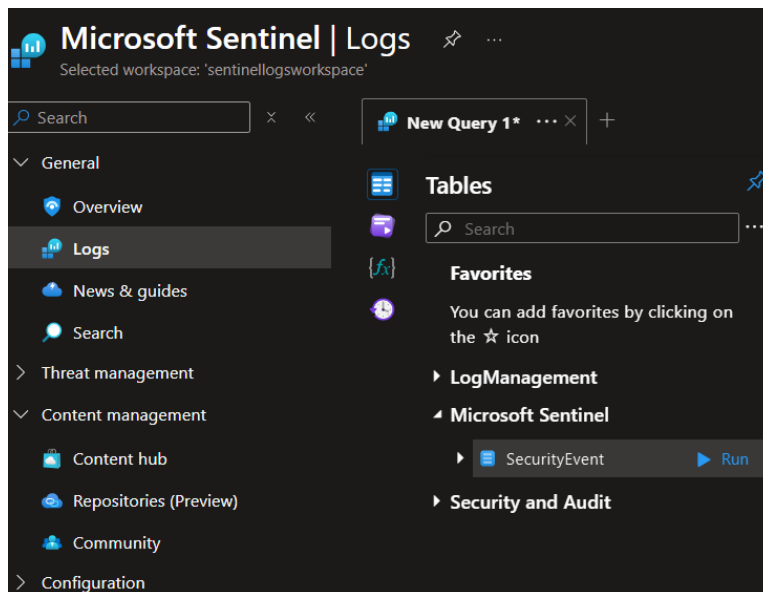
Date (UTC)	IOC	Malware	Tags	Reporter
2025-05-13 12:01:41	 79.110.49.229:7001	 Quasar RAT	AS399486 c2 censys quasar RAT VIRTUO	 DonPasci
2025-05-13 12:01:40	 23.94.99.5:5555	 Quasar RAT	AS-COLOCROSSING AS36352 c2 censys quasar RAT	 DonPasci
2025-05-13 08:01:39	 45.144.212.170:5938	 Quasar RAT	AS214940 c2 censys KPRONET quasar RAT	 DonPasci
2025-05-13 01:31:49	 eba378fab4635e2734cd38c33a75b...	 Quasar RAT		 Grim
2025-05-13 01:31:49	 be928f178a4ec59729f02200c6bd9...	 Quasar RAT		 Grim

Step 2: Prepare Your Environment for Threat Hunting in Azure Microsoft Sentinel

I will be using **Azure Microsoft Sentinel** for the threat hunt. Before starting the hunt, ensure that the Sentinel instance is set up and that you're ingesting logs into the **SecurityEvents** table.

1. **Log in to Azure Microsoft Sentinel:**
 - Go to [Azure Portal](#) and navigate to **Azure Microsoft Sentinel**.
2. **Check Data Availability:**
 - Make sure that the **SecurityEvents** table is collecting logs from your environment (e.g., Windows Server security logs or other log

sources). This is where you'll search for the IOCs. Open the logs database, in the left pane, under the general section, open Logs. Within the new query, choose Tables, then expand Microsoft Sentinel and click Run in the SecurityEvent selection.

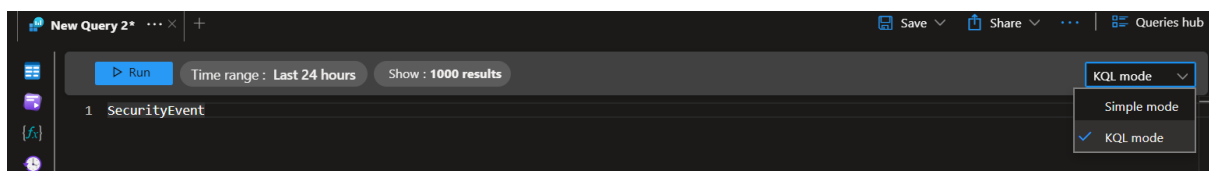


Step 3: Build a KQL Query to Search for IOCs

Now that you've gathered your IOCs, it's time to use **Kusto Query Language (KQL)** to search the **SecurityEvents** table for signs of compromise.

1. Access the Log Analytics Workspace

1. Open **Azure Microsoft Sentinel** in the portal.
2. In the **Threat Management** section, click **Logs** to open the query editor. On the right-hand side, click the dropdown arrow and choose KQL mode.



2. Write a KQL Query to Search for IOCs

- **Search for Malicious IP Addresses:** To check the database against suspicious IP addresses, type in the following command and click run.

Step 4: Run the Query and Analyse the Results

1. **Run the KQL Query:**
 - Click **Run** to execute the query and see the results.

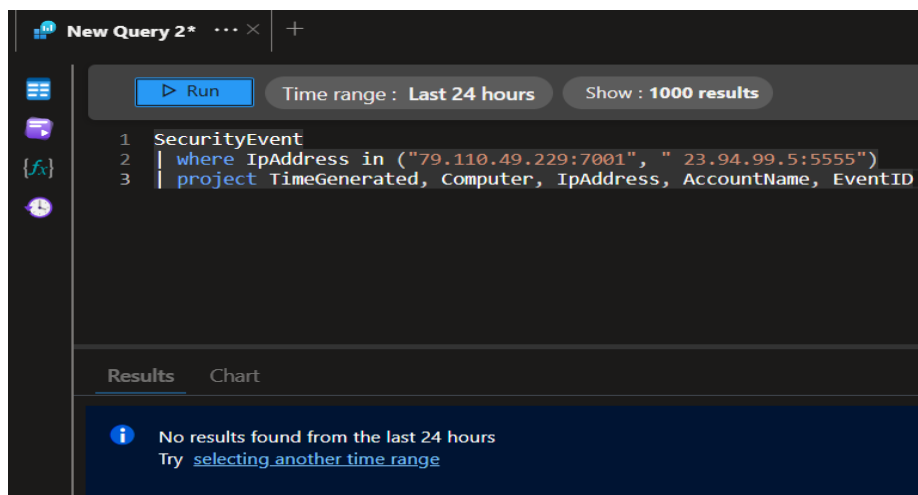
2. Analyze the Results:

- Review the logs for any matches to the IOCs you gathered.
- Investigate the event details such as the **TimeGenerated**, **Computer**, **AccountName**, and **CommandLine** information to determine the context of the potential threat.
- If matches are found, consider taking further steps to investigate or mitigate the threat.

SecurityEvent

| where IPAddress in ("[79.110.49.229:7001](#)", "[23.94.99.5:5555](#)")

| project TimeGenerated, Computer, IPAddress, AccountName, EventID



***There have been no results found in the database.

- **Search for Malicious Domains in CommandLine Logs:**

SecurityEvent

| where CommandLine has_any ("[dagodnox.ddns.net](#)", "[https://egirlcam.com/](#)")

| project TimeGenerated, Computer, CommandLine, AccountName

***There have been no results found in the database.

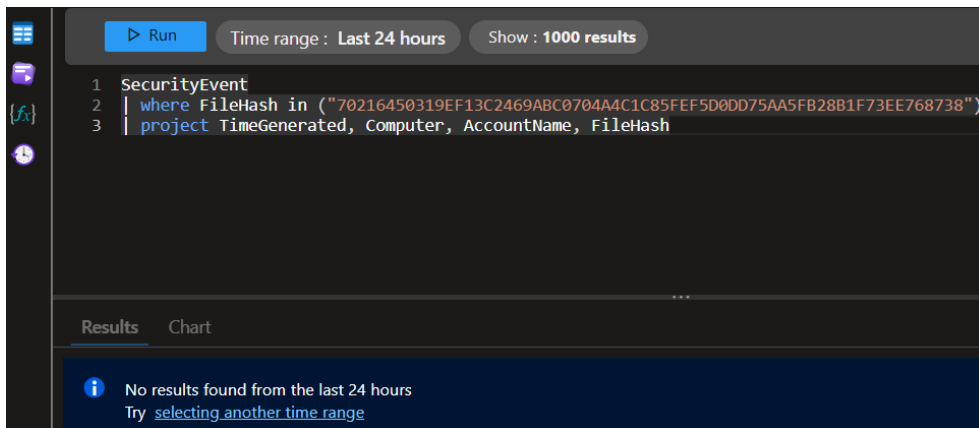
- **Search for Malicious File Hashes:**

SecurityEvent

| where FileHash in

("70216450319EF13C2469ABC0704A4C1C85FEF5D0DD75AA5FB28B1F73EE768738")

| project TimeGenerated, Computer, AccountName, FileHash

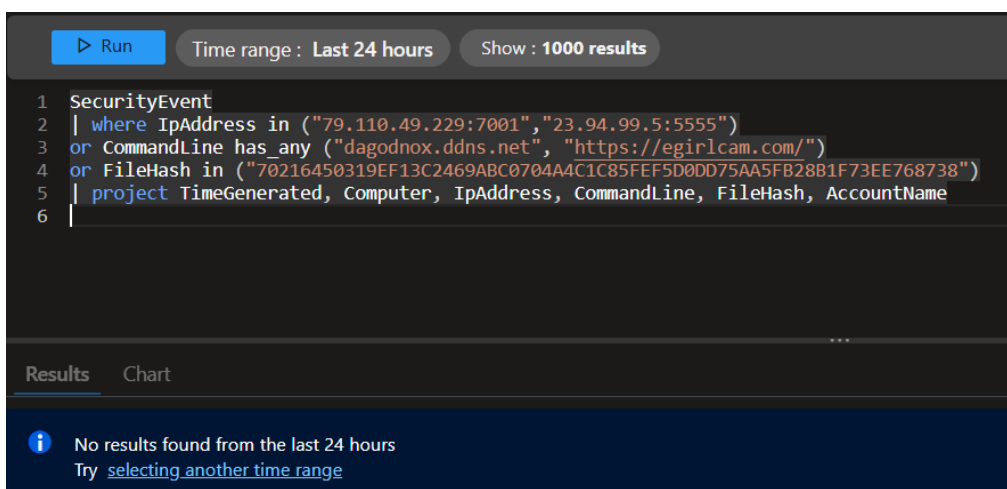


***There have been no results found in the database.

3. Combine IOCs into a Single Query (Optional)

***To run a broader search for all types of IOCs, you can combine them in one query:

```
SecurityEvent
| where IpAddress in ("79.110.49.229:7001","23.94.99.5:5555")
or CommandLine has_any ("dagodnox.ddns.net", "https://egirlcam.com/")
or FileHash in
("70216450319EF13C2469ABC0704A4C1C85FEF5D0DD75AA5FB28B1F73EE768738")
| project TimeGenerated, Computer, IpAddress, CommandLine, FileHash,
AccountName
```



***There have been no results found in the database.

Conclusion

In this project, I was able to apply core cybersecurity skills by integrating threat intelligence with proactive threat hunting techniques. I began by identifying reliable open-source threat intelligence platforms and gathering relevant Indicators of Compromise (IOCs), which are essential in building an effective threat hunting workflow.

Next, I gained hands-on experience with Kusto Query Language (KQL), learning how to write and refine queries to search and analyse log data in Azure Microsoft Sentinel. This enabled me to sift through large datasets efficiently and pinpoint suspicious activity.

Finally, I applied proactive threat hunting techniques by using the collected IOCs to search for signs of advanced threats within the SecurityEvents logs. This approach allowed me to uncover potentially hidden malicious activity, reinforcing the importance of continuous monitoring and analysis in a modern security operations environment.

Overall, this project helped bridge the gap between threat intelligence and detection, enhancing my ability to identify, analyse, and respond to evolving cybersecurity threats.